NPC – Unit 3: Configure a DNS Server and Web Server Using Cisco Packet Tracer

Practical Report – Networking Protocols & Communication (NPC)

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Subject	NPC – Unit 3
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Tool Used	Cisco Packet Tracer 9.0.1

1. Aim

To configure a DNS server and a web server in Cisco Packet Tracer so that a website can be accessed by its domain name (**www.npcunit3.com**) instead of its IP address.

2. Objectives

- To build a simple LAN containing two PCs, a switch, a DNS server and a web server.
- To assign static IP addresses to all devices.
- To host a web page on the web server over HTTP.
- To create a DNS A record that maps a domain name to the web server's IP address.
- To verify connectivity with ping and verify name resolution by browsing to the site from a PC.

3. Theory

DNS (Domain Name System) translates human-readable domain names into IP addresses. When a user types a name such as www.npcunit3.com into a browser, the client sends a query to its configured DNS server, which replies with the matching IP address recorded in an **A record**.

HTTP (Hypertext Transfer Protocol) is the application-layer protocol used to request and deliver web pages. Once the browser knows the server's IP address from DNS, it opens an HTTP connection to the web server and downloads the page.

A **Layer-2 switch** connects all devices within the same subnet, so no router or default gateway is needed when every device sits in a single network.

4. Network Overview

A single LAN, `192.168.1.0/24`, is built with two PCs, a Layer-2 switch, a DNS server and a web server. The web server hosts a simple page over HTTP, and the DNS server resolves **www.npcunit3.com** to the web server's address.

Devices: 2 × PC (PC0, PC1), 1 × Switch, 1 × DNS Server, 1 × Web Server.

[Insert logical topology diagram here – `logical_topology.png`]

[Insert Packet Tracer topology screenshot here – `screenshots/`]

5. IP Addressing Table

Device	IP Address	Subnet Mask
DNS Server	192.168.1.2	255.255.255.0
Web Server	192.168.1.3	255.255.255.0
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.11	255.255.255.0

- DNS server configured on the PCs: `192.168.1.2`
- DNS A record: `www.npcunit3.com` → `192.168.1.3`
- Default gateway: not required, because all devices are in a single subnet.

6. Procedure

Step 1: Build the topology

Place two PCs, one switch, and two servers in the Packet Tracer workspace. Connect each device to the switch using copper straight-through cables.

Step 2: Assign IP addresses

Open the **Desktop → IP Configuration** (or **Config → FastEthernet0**) tab of each device and set the static IP address and subnet mask from the addressing table. On PC0 and PC1, also set the **DNS Server** field to `192.168.1.2`.

Step 3: Configure the web server

On the Web Server, open **Services → HTTP**, make sure HTTP is turned **On**, and edit `index.html` to display a simple welcome page.

Step 4: Configure the DNS server

On the DNS Server, open **Services → DNS** and turn the DNS service **On**. Add a new record:

Field	Value
Name	www.npcunit3.com
Type	A Record
Address	192.168.1.3

Click **Add** to save the record.

Step 5: Test connectivity

From PC0, open the **Command Prompt** and run:

```
ping 192.168.1.2
ping 192.168.1.3
```

Step 6: Test name resolution and the website

On PC0, open the **Web Browser** and enter `http://www.npcunit3.com`. The page hosted on the web server should load.

7. Results

Test	Expected	Observed
Ping PC0 → DNS Server (192.168.1.2)	Replies received	4/4 replies, 0% packet loss
Ping PC0 → Web Server (192.168.1.3)	Replies received	4/4 replies, 0% packet loss
Browse `http://www.npcunit3.com` from PC0	Web page opens	Website opened successfully

[Insert ping test screenshot here]

[Insert website-in-browser screenshot here]

Both ping tests returned 4/4 replies with 0% packet loss, and the website opened successfully on PC0 via `http://www.npcunit3.com`.

8. Conclusion

A DNS server and a web server were successfully configured in Cisco Packet Tracer on a single LAN. The DNS A record resolved `www.npcunit3.com` to `192.168.1.3`, and the web page was reachable by name from the client PC. This confirms that DNS name resolution and HTTP access are working correctly.

9. Project Files

File	Description
`NPC_Unit_3_Shubham_Shinde.pdf`	Full practical report (submission copy)
`NPC_Unit_3_DNS_Web_Server.pkt`	Cisco Packet Tracer project file

File	Description
`screenshots/`	Packet Tracer screenshots (topology, ping, website)
`build_pdf.py`	Script used to generate the PDF report
`logical_topology.png`	Logical topology diagram used in the report